

1 Geometric Transformations

- Find the 3×3 matrix that corresponds to the composite transformation of a scaling by .3, a rotation of 90° about the origin, and finally a translation that adds $(-0.5, 2)$ to each point.
- Rotate points through 60° about the point $(6, 8)$.
- A camera in a 3D scene is positioned at $K(3, 4, 5)$. It first reflects across the yz -plane and then rotates 180° about the x -axis and finally translated by $(1, 4, 1)$. Find the new position of the point $(-1, 2, 3)$ after these transformations.
- Consider a 3D point $P = (6, 8, 12)$ in Cartesian coordinates. A perspective projection is applied using the projection plane $z = 4$ and a centre of projection (COP) at the origin $(0, 0, 6)$.
 - Compute the coordinates of the projected point P' on the projection plane.
 - Verify that the projected point lies on the plane $z = 4$.
- Let $\mathbf{u} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$
 - Find the Householder Matrix H that reflects each vector about a subspace W orthogonal to $\mathbf{u}$.
 - What will be the eigenvalues and the eigenvector of H .
 - What will be H^{101}
- Find the matrix of transformation that reflects each vector about the plane $x + 2y + z = 0$.
(**Hint:** First find the unit vector $\mathbf{u}$ orthogonal to given plane.)

2 Orthogonal Projections & QR- Decomposition

7. Let

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ -1 & 1 \end{bmatrix} \quad \mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

- Find the orthogonal complement of the column space of A . **Or** Find all vectors orthogonal to the columns of A . **Or** Find $(\text{col}(A))^\perp$.
- Find the projection of $\mathbf{u}$ on $\text{Col}(A)$.
- Find the orthogonal decomposition of u w.r.t $\text{Col}(A)$ **Or** Write $\mathbf{u}$ as sum of two components one belongs to $\text{Col}(A)$ and other in $(\text{col}(A))^\perp$.
- Extend the $\text{Col}(A)$ to form the orthogonal basis of $\mathbb{R}^3$.
- Find a QR factorization of A , where Q is orthogonal matrix and R is upper triangular matrix.

3 Least Square Solution

8. Let

$$A = \begin{bmatrix} 1 & 3 \\ 1 & -1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 1 \\ 0 \end{bmatrix}$$

- (a) Find a least-squares solution of $A\mathbf{x} = \mathbf{b}$.
 - (b) Find the error vector $\mathbf{e}$.
 - (c) Find the Left inverse of A .
 - (d) Use left inverse to find least square solution of $A\mathbf{x} = \mathbf{b}$.
 - (e) If $A = QR$, what will be the expression for the left inverse of A .
 - (f) Use left inverse of A to find the projection matrix that projects on to the $Col(A)$.
 - (g) find the projection of b onto the $Col(A)$.
 - (h) Find the best fit line $y = a + bx$ for the data $(3, 5)$, $(-1, 1)$, $(1, 0)$.
9. Define an inner product on $\mathbb{P}_2(x)$ as

$$\langle \mathbf{p}, \mathbf{q} \rangle = \mathbf{p}(-1)\mathbf{q}(-1) + \mathbf{p}(0)\mathbf{q}(0)$$

- (a) Find the norm of x and x^2 w.r.t this inner product.
- (b) Find the projection of x on to the $span\{1, x^2\}$.
- (c) Find the orthogonal basis of $\mathbb{P}_2(x)$ w.r.t defined inner product.

4 Othogonal Diagonalization

10. For the matrix:

$$A = \begin{bmatrix} 6 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix},$$

- (a) Find the eigenvalues and corresponding eigenvectors.
- (b) Construct the orthogonal matrix Q and the diagonal matrix D .
- (c) Verify that $Q^T A Q = D$.

5 Quadratic Forms and Constraint Optimization

11. Consider the quadratic form

$$Q(x, y, z) = 6x^2 + 3y^2 + 5z^2 + 4xy$$

- (a) Write the quadratic form in matrix notation $Q(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$, where A is a symmetric matrix .
- (b) Determine whether the quadratic form is positive definite, negative definite, or indefinite. Justify your answer by examining the eigenvalues of the associated matrix.
- (c) Find the orthogonal transformation that diagonalizes the associated matrix. Express the quadratic form in terms of the new variables.

- (d) Describe the geometric shape of level surface $Q(x, y, z) = 1$.
- (e) Find the maximum and minimum values of the given quadratic form subject to the constraint $x^2 + y^2 + z^2 = 1$, and determine the values of x , y and z at which the maximum and minimum occur.
- (f) Explain the role of positive/negative definiteness in ensuring the existence of a unique minimum/maximum for $Q(x, y, z)$.

6 Singular Value Decomposition

12. Let

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ -1 & 1 \end{bmatrix}$$

- (a) Find the Singular Value Decomposition(SVD) of the matrix A .
- (b) Compute the Spectral decomposition of the matrix A or write A as sum of rank 1 matrices.
- (c) Determine the orthogonal basis for the four fundamental subspaces of A .
- (d) What will be the Singular Value Decomposition(SVD) of the matrix A^T .
- (e) Compute the left inverse of the matrix A using its SVD.
- (f) Find the least square solution of $Ax = b$
- (g) Determine the projection matrix that projects onto $col(A)$

7 Principal Component Analysis

13. Given the covariance matrix:

$$\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix},$$

perform the following tasks:

- (a) **Eigenvalues and Eigenvectors:** Compute the eigenvalues and eigenvectors of the covariance matrix Σ .
 - (b) **Principal Directions:** Identify the principal directions (principal components) of the data.
 - (c) **Variance Explained:** Compute the proportion of total variance explained by each principal component.
 - (d) **Dimensionality Reduction:** Assume the data points are transformed using only the first principal component. How much variance is retained?
14. Consider the following dataset with two features X_1 and X_2 , observed across four data points:

Observation	X_1	X_2
1	2	4
2	4	2
3	6	8
4	8	6

Perform the following tasks:

- (a) Compute the Centralized matrix.
- (b) Compute the covariance matrix.
- (c) Find the eigenvalues and eigenvectors of the covariance matrix.
- (d) Determine how much variance is preserved when projecting the data onto the first principal component.
- (e) Transform the data along the first principal component.
- (f) Project the data onto the first principal component.

8 Chapter # 8

15. Let

$$\mathbf{v}_1 = \begin{pmatrix} 3 \\ -3 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 0 \\ 6 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad \text{and} \quad \mathbf{p} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

- (a) Check affine independence of the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, if so find barycentric coordinates of $\mathbf{p}$ with respect to given affinely independent set, if set is affinely dependent give dependence relation.
- (b) $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$
- (c) $\text{aff}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$
- (d) $\text{conv}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$

9 Tensor Unfolding

16. Given the tensor $\mathcal{A} \in \mathbb{R}^{2 \times 3 \times 2}$:

$$\mathcal{A}(:, :, 1) = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad \mathcal{A}(:, :, 2) = \begin{bmatrix} 7 & 8 & 9 \\ 10 & 11 & 12 \end{bmatrix}.$$

- (a) Unfold $\mathcal{A}$ along **mode-1**, and write the resulting matrix $A_{(1)}$.
- (b) For the same tensor $\mathcal{A}$, unfold it along **mode-2**, and write the resulting matrix $A_{(2)}$.
- (c) Unfold the tensor $\mathcal{A}$ along **mode-3**, and write the resulting matrix $A_{(3)}$.

17. For a tensor $\mathcal{B} \in \mathbb{R}^{4 \times 3 \times 5}$, write the dimensions of $B_{(1)}$, $B_{(2)}$, and $B_{(3)}$.

10 Mode-n Product

18. **Mode-1 Product** Given $\mathcal{A} \in \mathbb{R}^{2 \times 3 \times 2}$ as in and a matrix $M \in \mathbb{R}^{4 \times 2}$:

$$M = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Compute $\mathcal{B} = \mathcal{A} \times_1 M$, and provide the slices $\mathcal{B}(:, :, 1)$ and $\mathcal{B}(:, :, 2)$.

19. **Mode-2 Product** For the same tensor $\mathcal{A}$, compute $\mathcal{B} = \mathcal{A} \times_2 N$, where:

$$N = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 1 & 1 \end{bmatrix}.$$

20. **Mode-3 Product** For the tensor $\mathcal{A}$ and matrix $P \in \mathbb{R}^{3 \times 2}$:

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix},$$

compute $\mathcal{B} = \mathcal{A} \times_3 P$, and describe the resulting tensor's dimensions.

11 Higher Order Singular Value Decomposition

21. For the Given Tensor: $\mathcal{A} \in \mathbb{R}^{1 \times 2 \times 3}$

$$\mathcal{A}(:, :, 1) = \begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \mathcal{A}(:, :, 2) = \begin{bmatrix} 1 & -1 \end{bmatrix}, \quad \mathcal{A}(:, :, 3) = \begin{bmatrix} 0 & 1 \end{bmatrix}$$

Compute the HOSVD and find:

- Core tensor $\mathcal{S}$,
- Factor matrices $U^{(1)}$, $U^{(2)}$, and $U^{(3)}$.